Haider Ghanem

Full-Stack Engineer (Python, React, Flutter, Machine Learning)

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SUMMARY

Full-stack engineer who designs and builds complete products: Python backends with FastAPI and Django, React and Flutter front ends, and the data and machine learning work behind them. 8+ production systems shipped, with a focus on security, multi-tenant design and real-time features. Open to remote and on-site work.

SKILLS

Backend: Python, FastAPI, Django, Django REST Framework, Django Channels, Flask, Pydantic, Node.js, Express, WebSockets, Celery, REST APIs, OpenAPI

Frontend: React, TypeScript, JavaScript, Next.js, Tailwind CSS, Three.js, React Three Fiber, Zustand, TanStack Query, Vite, PWA, Web Audio, Web MIDI

Mobile: Flutter, Dart, Riverpod, GoRouter, Dio, Firebase Cloud Messaging

AI and LLMs: OpenAI API, Claude API, Gemini API, Faster-Whisper, TensorFlow Lite, LLM integration

Data and machine learning: Pandas, NumPy, SciPy, Scikit-learn, XGBoost, SMOTE, Matplotlib, Seaborn, Plotly, Jupyter, Streamlit

Databases: PostgreSQL, SQLite, SQLAlchemy, Alembic, Redis, Firebase (Auth, Firestore), Supabase, MinIO / S3

DevOps: Linux, Git, Docker, Docker Compose, GitHub Actions (CI/CD), Nginx, AWS S3, Prometheus

Testing: pytest, Vitest, Playwright, Flutter test, CI on every push

Security: RBAC, JWT, two-factor authentication (TOTP), rate limiting, audit logs, multi-tenant isolation, argon2 / bcrypt password hashing, container isolation

Integrations: Stripe, Twilio (SMS, WhatsApp), Telegram Bot API (aiogram), Google Maps, Leaflet

Networking: Cisco routing and switching (CCNA labs in Packet Tracer)

Other: Arabic and English interfaces (RTL, i18n), technical documentation (SRS, ERD, architecture)

PROJECTS

Smart Ordering Platform - multi-tenant Telegram ordering SaaS

Django, DRF, Channels, Celery, aiogram, PostgreSQL, React, TypeScript | github.com/h19kod/telegram_bot

- Lets restaurants and shops in Iraq take orders through a Telegram bot and Mini App, managed from a web dashboard, with an optional AI ordering assistant in Iraqi Arabic.
- Strict tenant isolation: every query is scoped to its business and covered by an isolation test matrix.
- Server-side re-pricing, Telegram initData validation, and idempotent checkout with optimistic locking.

VoxLingo AI - AI video subtitling and dubbing

FastAPI, Celery, PostgreSQL, Redis, S3, FFmpeg, React | github.com/h19kod/tool_ai

- Transcribes videos with Faster-Whisper and translates them with Claude, using a glossary and translation memory.

- Dubs with a voice per speaker, fitted to the original timing, and produces correct right-to-left Arabic subtitles.
- Resumable background jobs with live progress over server-sent events.

Diabetes Prediction - clinical risk model

Python, Pandas, Scikit-learn, XGBoost, SMOTE | github.com/h19kod/data_anlysis_diabitc

- Tuned model that flags diabetes risk from 14 clinical indicators: 92.3% accuracy, 0.956 ROC-AUC, 84.6% recall.
- SMOTE class balancing raised recall by 23.1 points over the baseline model.

TripMind AI - AI travel planner (Flutter)

Flutter, Dart, Riverpod, GoRouter, Firebase, OpenAI API | github.com/h19kod/Ai_travelsystem_app_flutter

- 30-screen app that turns a destination, budget and dates into a day-by-day itinerary, with an AI chat assistant that falls back to a built-in guide offline.
- Firebase Auth and Firestore, full Arabic and English support, with tests and CI on every push.

PIANOVA - interactive piano platform

Next.js, TypeScript, Supabase, Three.js, Vitest, Playwright | github.com/h19kod/pianova

- 88-key sampled piano in the browser, played by mouse, multi-touch, computer keyboard or Web MIDI.
- Composition studio with piano roll and MIDI import, a 3D piano, and offline support as a PWA.

LANGUAGES

Arabic: native. English: professional working proficiency.